

SPBANKLENS 360

Enterprise Banking Operations & Risk Intelligence Platform

BUSINESS REQUIREMENTS DOCUMENT (BRD)

Version 1.0 Prototype

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Version History

Version	Date	Author	Summary of Changes
0.1	20 Jun 2026	Sagarika Paul	Initial draft outline aligned to product vision
0.9	24 Jun 2026	Sagarika Paul	Module-level requirements captured from prototype build
1.0	27 Jun 2026	Sagarika Paul	Final BRD baselined against implemented prototype

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1. Executive Summary

SPBANKLENS 360 is an enterprise-grade Risk Intelligence and Operational Monitoring platform designed to give banks a unified, role-based operational view across Branches, Regional Offices, and Central Office. The product is implemented as a high-fidelity prototype (Version 1.0) that demonstrates how fragmented banking operations data — loan reviews, recovery follow-ups, inspection observations, legal cases, CGTMSE renewals, KYC and documentation gaps, early-warning signals, and officer task queues — can be consolidated into a single, read-only operational intelligence layer.

The platform is intentionally architected as a non-transactional overlay. It does not replace the Core Banking System (CBS), Loan Origination System (LOS), or any system of record. Instead, it aggregates operational signals from these systems and presents them through dashboards, work queues, knowledge notebooks, escalation matrices and exception reports — calibrated to the daily decisions made at each level of the bank.

This Business Requirements Document captures the business needs, scope, stakeholders, processes, rules, and benefits realised by the implemented prototype, and clearly distinguishes future enhancements from delivered capability.

2. Business Background

Indian and global banking operations have grown more complex over the last decade: regulatory expectations have sharpened, asset quality scrutiny has intensified, and customers expect faster service. Branches, however, continue to operate with a patchwork of operational tools — CBS for transactions, LOS for origination, multiple Excel trackers for everything else, and tacit institutional knowledge that lives in the heads of individual officers.

Common operational pain points observed across mid-sized to large banks include:

- Fragmented operational data spread across CBS, LOS, KYC systems, CGTMSE portals, legal trackers, inspection registers, HRMS and external bureau feeds.
- Excessive reliance on Excel trackers — separate workbooks for SMA, NPA, recovery, documentation, insurance, valuation, CGTMSE, legal and inspection.
- Officer transfer issues, where the institutional memory of a borrower (visits, conversations, recovery efforts, family situation, business context) is lost when officers move.
- Manual follow-up routines that depend on diary entries, post-its and personal calendars, leading to missed reminders and SLA breaches.
- Inspection challenges — scattered evidence, late closure of observations, and limited visibility for Regional Office on inspection readiness across branches.
- Recovery tracking that lacks structured visibility into action history, OTS discussions, SARFAESI timelines, DRT hearings and visit notes.
- Legal monitoring spread across legal cell registers, advocate letters and email threads, without a single chronological view.
- Compliance deadlines (KYC review, insurance renewal, valuation, CGTMSE renewal) that surface only after they slip.
- Lack of portfolio visibility for Branch Managers, Regional Heads and Central Office to compare branches, regions, sectors and risk bands at a glance.

3. Business Problem Statement

Bank branches operate critical risk decisions on incomplete, late, and personally-held information. There is no single operational layer that:

- Brings together all operational signals for a borrower in one Account 360 view.
- Preserves institutional memory across officer transfers via a structured Knowledge Notebook.
- Generates and reminds officers of operational tasks based on rule-based risk triggers.
- Gives supervisors (Branch Manager, Regional Office, Central Office) calibrated, role-appropriate views of branch health, escalations and exceptions.
- Allows operational data to be reviewed, monitored and reported on without writing back to the CBS or LOS.

The absence of such a layer manifests as missed reviews, delayed recovery action, repeat audit observations, weak handover during transfers, and limited early warning on deteriorating accounts.

4. Vision Statement

SPBANKLENS 360 envisions a unified Operational Intelligence Layer that sits transparently above all banking systems and serves as the daily operating cockpit for every officer, manager, and executive in the bank.

The long-term vision is a platform where:

- Branches operate from a single screen showing what is due, what is overdue, and what is high-risk — for every borrower.
- Regional Offices supervise branches with comparable, ranked, and heat-mapped intelligence rather than ad-hoc Excel returns.
- Central Office gains an enterprise-wide command centre view of risk concentration, governance health, and inspection posture.
- Institutional memory accumulates over time as a versioned, searchable, structured Knowledge Notebook for every account.
- Every dashboard, alert and report is read-only, role-based, and auditable — preserving the integrity of the systems of record.

5. Business Objectives

Objective	Description	Measurable Outcome
Operational visibility	Single role-based dashboard for Branch, RO and CO	100% of operational KPIs visible in ≤ 2 clicks
Officer continuity	Structured handover notebook for borrower history	Reduce knowledge loss during transfers
Reduce manual tracking	Eliminate parallel Excel trackers for SMA, NPA, recovery, legal	Consolidate ≥ 8 trackers into 1 platform
Inspection readiness	Central register of observations, ATR status and evidence	Faster closure of audit findings
Strengthen governance	Enterprise heat maps, governance indices, threshold controls	Earlier identification of weak branches/regions
Improve recovery monitoring	Structured recovery actions, OTS, SARFAESI, DRT tracking	Higher recovery rate and on-time action
Centralize operational info	Read-only intelligence layer with audit trail	Single source of operational truth

6. Scope

6.1 In Scope (Implemented in Prototype v1.0)

The following modules are implemented and demonstrable in the prototype:

Module	Implemented Capability
Application Shell & Security	Enterprise navigation, sidebar, top bar, search, notifications, profile; role-based login (Branch, Regional, Central, Admin); protected routes; encrypted server-side session cookies.
Branch Dashboard	16 operational KPIs, Risk Health Index, portfolio trends, risk distribution, sector exposure for the branch.
Account 360	Single-screen borrower view with risk dial, early warning panel, interaction history, exposure and security.
Knowledge Notebook 2.0	Structured 10-section operational notes (Status, Interaction, Business Obs, Recovery, Legal, Inspection, Risk, Action, Attachments) with version history, global search, quick-note capture.
Officer Continuity	Handover register tracking pending actions, sensitive borrowers and follow-ups during officer transfers.
Regional Office Dashboard	Branch monitoring, risk heat map, branch ranking, comparison tools, regional follow-ups and escalations, Excel exports.
Central Office Command Centre	National Health Index, regional risk heat maps, sector and geographic concentration analytics, governance and ATR trackers, admin thresholds.
Risk Intelligence Module	Data Integration Hub (Excel/CSV upload with validation), Rule-Based Risk Engine (13 weighted rules), Early Warning System, Task Generation, Exception Management.
Workflow Centre	Officer Work Queue, Manager Workbench, Operational Calendar, SLA dashboard, Reminder Engine, Email Automation simulator, Escalation Matrix, Notifications.
NPA Lifecycle	Standard → SMA-0/1/2 → NPA stage tracking, ageing buckets, provisioning view.
Legal Register	Litigation register, SARFAESI 13(2)/13(4), DRT hearings, limitation calendar.
CGTMSE Management	Registration, fees, renewals, claims, queries, settlements with calendar reminders.
Inspection & Audit	Inspection register, observation tracking, ATR (Action Taken Report) monitoring.
Executive War Room	Large borrower concentration, sector exposure, systemic risk analytics.
MIS / Reports	Daily, weekly, monthly, quarterly executive reports with PDF/Excel export and product-branded watermarks.

Module	Implemented Capability
Product Vision & About	Public information pages explaining product purpose and developer credits.

6.2 Out of Scope (Future Enhancements — Not Implemented in v1.0)

- Live API integration with CBS, LOS, KYC, CGTMSE portal, legal and inspection systems.
- Enterprise SSO (LDAP/AD/SAML/Okta) authentication.
- Native mobile applications (iOS / Android).
- Machine learning models for predictive risk scoring.
- Automated outbound emails/SMS to customers.
- Write-back of any data to source systems.
- Multi-bank / multi-tenant deployment.
- Production-grade encryption-at-rest and HSM-based key management.

7. Stakeholders

Stakeholder	Interest in Platform
Branch Officers	Daily work queue, account-level follow-up, knowledge notes
Branch Managers	Branch health, officer performance, escalations
Regional Office	Branch comparison, heat map, escalations, regional reports
Central Office	National health index, sector/geo concentration, governance
Operational Risk Teams	Risk thresholds, rule weights, EWS calibration
Inspection Teams	Observation tracking, ATR status, branch readiness
Recovery Teams	Recovery action history, OTS, SARFAESI, DRT timelines
Compliance Teams	KYC, insurance, valuation, CGTMSE compliance posture
IT Support	Application availability, integration roadmap
System Administrators	RBAC, thresholds, escalation SLAs, audit configuration

8. Business Users

8.1 Branch User (Officer / Manager)

Primary operational user. Uses the Branch Dashboard, Account 360, Knowledge Notebook, Task Queue, Calendar and module screens (Loan Reviews, Documentation, Recovery, Legal, CGTMSE, Inspection) to manage day-to-day work.

8.2 Regional User

Supervisory user. Monitors all branches under a region through dashboards, heat maps, branch comparison, follow-ups and escalation matrix; consumes regional Excel reports.

8.3 Central User

Executive user. Operates the enterprise command centre, governance trackers, war room analytics, audit/inspection oversight, and configures bank-wide thresholds and SLAs.

8.4 Security / Admin User

Administers role assignments, threshold configuration, escalation SLAs, and reviews the audit trail of metadata changes and interactions.

9. Business Processes

Process	End-to-End Flow in SPBANKLENS 360
Loan monitoring	Loan portfolio is pulled into the Branch Dashboard; SMA/NPA classification is shown; high-risk accounts surface in EWS; officer drills into Account 360; observations logged in Knowledge Notebook; review tasks auto-generated.
Recovery	Recovery Queue surfaces overdue accounts; officer records visit/contact in Knowledge Notebook (Recovery section); legal milestones (SARFAESI 13(2), 13(4), DRT) tracked in Legal Register; OTS discussions captured as notes; reminders fire from the Reminder Engine.
Inspection	Inspection register holds observations from statutory/internal audits; ATR status updated; calendar reminders for closure; Central Office dashboard tracks open observations by branch and region.
Legal	Litigation Register tracks notices issued, hearings, advocate assignments and limitation dates; calendar alerts trigger ahead of cut-off dates.
Documentation	Documentation gaps (KYC, insurance, valuation, CGTMSE renewal) appear as KPI tiles and tasks; closure flows back into compliance score.
Workflow management	Tasks are generated by the Rule Engine and EWS; appear in Officer Work Queue and Manager Workbench; SLA dashboard tracks ageing; escalation matrix promotes overdue items to higher levels.
Officer handover	Outgoing officer's pending actions, sensitive borrowers and notebook entries are presented to the incoming officer through the Officer Continuity module.
Risk monitoring	Rule engine combines 13 weighted indicators; EWS raises alerts; risk score is explainable via the 'Explainable Risk' panel; thresholds are admin-configurable.
Reporting	Daily/weekly/monthly/quarterly reports generated for each level; exports to PDF/Excel with product-branded watermarks; regional Excel pack generated via xlsx engine.

Note: System architecture and data flow are illustrated in the companion diagrams 'SPBANKLENS_360_Architecture' and 'SPBANKLENS_360_DataFlow'.

10. Business Requirements

10.1 Application Shell & Security

ID	Requirement	Priority
BR-SH-01	Provide a unified application shell with sidebar, top bar, global search, notifications and user profile.	Must
BR-SH-02	Support four roles — Branch, Regional, Central, Admin — with role-based routing.	Must
BR-SH-03	Authenticate via encrypted httpOnly server-side session cookies; no auth tokens in browser storage.	Must
BR-SH-04	Enforce protected routes that redirect unauthorised users to login.	Must

10.2 Branch Dashboard

ID	Requirement	Priority
BR-BR-01	Display 16 operational KPIs (Standard, SMA-0/1/2, NPA, overdue, reviews, documentation, KYC, etc.).	Must
BR-BR-02	Provide Risk Health Index for the branch.	Must
BR-BR-03	Visualise portfolio trend, risk distribution and sector exposure.	Must
BR-BR-04	Surface high-risk accounts with one-click drill into Account 360.	Must

10.3 Account 360 & Knowledge Notebook

ID	Requirement	Priority
BR-A3-01	Show borrower profile, exposure, security, risk dial and early warnings on a single screen.	Must
BR-A3-02	Allow structured notebook entries across 10 sections (Status, Interaction, Business, Recovery, Legal, Inspection, Risk, Action, Attachments, Remarks).	Must
BR-A3-03	Maintain version history for every notebook entry.	Must
BR-A3-04	Support global search across all notebook entries.	Must
BR-A3-05	Provide a Quick Note dialog from list rows and Latest Note cards.	Should

10.4 Officer Continuity

ID	Requirement	Priority
BR-OC-01	Track pending actions, sensitive borrowers and unresolved notes for officer transfers.	Must
BR-OC-02	Present a structured handover view to incoming officers.	Must

10.5 Regional Office

ID	Requirement	Priority
BR-RO-01	Display regional KPIs and a Branch Risk Heat Map.	Must
BR-RO-02	Rank branches and compare any two side-by-side.	Must
BR-RO-03	Maintain RO follow-up and escalation register with status workflow.	Must
BR-RO-04	Generate regional Excel reports.	Must

10.6 Central Office

ID	Requirement	Priority
BR-CO-01	Provide Executive Command Centre with National Health Index.	Must
BR-CO-02	Render national/regional heat maps and sector/geo concentration analytics.	Must
BR-CO-03	Maintain Governance Index and ATR tracker.	Must
BR-CO-04	Allow admin to configure risk thresholds and escalation SLAs bank-wide.	Must

10.7 Risk Intelligence Module

ID	Requirement	Priority
BR-RI-01	Provide a Data Integration Hub for Excel/CSV uploads with validation.	Must
BR-RI-02	Run a Rule-Based Risk Engine with 13 configurable weighted rules.	Must
BR-RI-03	Produce an Explainable Risk panel showing why a score changed.	Must
BR-	Drive Early Warning System and auto-generate tasks on rule breach.	Must

ID	Requirement	Priority
RI-04		
BR-RI-05	Provide an Exception Management view.	Should

10.8 Workflow Centre

ID	Requirement	Priority
BR-WF-01	Provide Officer Work Queue and Manager Workbench tabs.	Must
BR-WF-02	Drive a Reminder Engine with windowed reminders (T-30, due today, overdue).	Must
BR-WF-03	Provide an Email Automation simulator for outbound notifications.	Should
BR-WF-04	Enforce an Escalation Matrix (Officer → Branch Manager → Regional Office).	Must
BR-WF-05	Render an Operational Calendar across all task categories.	Must
BR-WF-06	Show SLA dashboard with ageing and breach indicators.	Must

10.9 NPA, Legal, CGTMSE, Inspection, War Room

ID	Requirement	Priority
BR-NP-01	Track NPA lifecycle stages, ageing buckets and provisioning view.	Must
BR-LG-01	Maintain legal register with SARFAESI/DRT timelines and limitation calendar.	Must
BR-CG-01	Track CGTMSE registration, fees, renewals, claims and settlements.	Must
BR-IN-01	Track inspection observations and ATR closure status.	Must
BR-WR-01	Provide Executive War Room with concentration and systemic risk analytics.	Should

10.10 Reporting & Branding

ID	Requirement	Priority
BR-RP-01	Generate daily, weekly, monthly and quarterly MIS reports.	Must
BR-	Export to PDF, Excel and CSV with product-branded watermark.	Must

ID	Requirement	Priority
RP-02		
BR- RP-03	Display product name, version, developer credit and copyright across all surfaces.	Must
BR- RP-04	Show a one-time Prototype Disclaimer modal on first visit.	Must

11. Business Rules

1. Only authenticated users with a valid server-side session may access protected routes; the session cookie is httpOnly, Secure and SameSite=None.
2. Role-based access: Branch users see only branch routes; Regional users see regional routes; Central users see central routes; Admin sees configuration screens.
3. The platform is read-only — no module writes back to CBS, LOS or any system of record.
4. Risk score is computed by weighted aggregation of 13 rule outputs; weights are admin-configurable and every score change must be explainable.
5. Reminder Engine fires at three windows: T-30 days (advance), Due today, and Overdue (with day count).
6. Escalation Matrix auto-promotes overdue items through Officer → Branch Manager → Regional Office, based on configured SLA days.
7. Knowledge Notebook entries are immutable once saved; edits create a new version, preserving the original.
8. Reports and exports carry product branding and the 'Prototype Edition' watermark; cannot be edited from within the application.
9. All sample data is fictional unless manually uploaded by the user; uploaded data persists only in the browser session.

12. Expected Business Benefits

Benefit	Description
Operational efficiency	Single screen replaces multiple Excel trackers and ad-hoc emails
Better governance	Heat maps and indices give supervisors objective comparators
Portfolio visibility	Sector, geography, product and band-wise views at every level
Officer continuity	Structured notebook and handover module reduce knowledge loss
Faster issue identification	EWS, reminders and exception views surface issues earlier
Audit preparedness	Inspection register, ATR tracker and audit trail support audits
Consistent reporting	Branded MIS exports across daily/weekly/monthly cadences

13. Assumptions

- Source banking systems (CBS, LOS, KYC, CGTMSE, Legal, Inspection, HRMS) can, in a production rollout, expose data via batch files or APIs.
- Users have a modern browser and basic familiarity with web dashboards.
- Operational definitions (SMA, NPA, SARFAESI stages, ATR) are aligned to the bank's policy and regulator guidance.
- Sample/demo data sets used in the prototype are representative of typical branch composition but are fictional.
- Branding, developer credit and copyright remain intact across deployments.

14. Constraints

- Demo environment — runs as a hosted prototype, not a production-hardened bank deployment.
- Sample data — no real customer, account or transaction data is used or stored.
- No production banking integration — there are no live connections to CBS, LOS or any external regulator/credit bureau.
- Read-only upload architecture — data enters only via manual Excel/CSV upload; the platform never writes back to source systems.
- No native mobile app; the product is delivered as a responsive web application.
- Authentication is a prototype implementation suitable for demonstration; production rollout would require enterprise SSO.

15. Risks

Risk	Category	Mitigation
Misinterpretation of prototype as a production system	Business	Persistent Prototype Disclaimer modal and watermark on exports
Data privacy concerns if real data uploaded	Compliance	Read-only browser-session persistence; clear messaging that uploaded data is for demonstration only
Variations in operational definitions across banks	Functional	Admin-configurable thresholds and rule weights
User adoption resistance vs. existing Excel trackers	Change Mgmt	Familiar dashboard patterns and Excel-style exports
Future integration complexity with CBS/LOS	Technical	Read-only integration hub designed as a thin, replaceable layer
Loss of institutional memory before adoption	Operational	Knowledge Notebook designed for incremental capture from day one

16. Success Criteria

- All implemented modules are demonstrable end-to-end without runtime errors.
- Each user role logs in and reaches its intended dashboard within two clicks.
- Risk score, EWS, reminders and escalations behave consistently with the documented rules.
- Reports export with branded PDF/Excel and consistent typography.
- Knowledge Notebook persists entries with version history across navigation.
- Stakeholder review confirms that the prototype credibly represents the target operating model.

17. Future Roadmap

The items below are future possibilities and are NOT part of v1.0:

- Secure enterprise authentication — SSO via LDAP/AD/SAML/Okta with MFA.
- API integration with CBS, LOS, KYC, CGTMSE, legal and inspection systems (subject to bank approvals and InfoSec review).
- Native mobile application for officers and managers.
- Advanced analytics — cohort analysis, vintage curves, behavioural scoring.
- AI-assisted insights with human oversight (e.g. note summarisation, anomaly hints), with explainability and audit logging.
- Multi-bank/multi-tenant SaaS deployment with isolation guarantees.
- Workflow approvals and electronic sign-offs.

18. Appendices

18.1 Glossary

Term	Definition
ATR	Action Taken Report — response to an audit/inspection observation
BRD	Business Requirements Document
CBS	Core Banking System
CGTMSE	Credit Guarantee Fund Trust for Micro and Small Enterprises
CO	Central Office
DRT	Debt Recovery Tribunal
EB-RIL	Enterprise Banking — Risk Intelligence Layer (SPBANKLENS 360)
EWS	Early Warning System
KYC	Know Your Customer
LOS	Loan Origination System
MIS	Management Information System / Reports
NPA	Non-Performing Asset
OTS	One-Time Settlement
RBAC	Role-Based Access Control
RO	Regional Office
SARFAESI	Securitisation and Reconstruction of Financial Assets and Enforcement of Security Interest Act
SLA	Service Level Agreement
SMA	Special Mention Account (SMA-0/1/2)
SSO	Single Sign-On

18.2 Acronyms

All acronyms used in this document are expanded in the Glossary above.

18.3 References

- SPBANKLENS 360 Software Requirements Specification (SRS) v1.1
- SPBANKLENS_360_Architecture.png — System architecture diagram
- SPBANKLENS_360_DataFlow.png — Data flow diagram
- SPBANKLENS 360 Product Vision page (/product-vision) and About page (/about) within the application

18.4 Version History

Refer to the Version History page at the start of this document.